

Machine Learning for Historical Groundwater Arsenic Risk Assessment in Bangladesh: Spatial Validation, Explainability, and Uncertainty

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Abstract. Arsenic contamination of shallow tube wells is one of the largest mass poisoning events in history, affecting tens of millions of people in Bangladesh, yet most wells have never been tested and laboratory analysis is slow and costly. This study asks whether machine learning can extract reliable, explainable risk patterns from the 1998–1999 National Hydrochemical Survey of Bangladesh ($N = 3,534$ wells), while being explicit that the survey describes historical conditions only. Exceedance of the national arsenic standard ($50 \mu\text{g/L}$; 24.9% of wells) was modeled with five algorithms – random forest, XGBoost, LightGBM, CatBoost, and a neural network (MLP) – benchmarked against dummy and logistic-regression baselines, and evaluated under random, single-spatial-block, and eight-times-repeated spatial-block holdouts. The gradient-boosted and random-forest models reached mean ROC-AUC of 0.950–0.953 under repeated spatial holdout ($SD = 0.008$ – 0.009), with random splits not inflating scores (mean difference = 0.001). SHAP attribution identified phosphorus, iron, and strontium as the variables the best model relied on most. Conformalized quantile regression produced 90% intervals covering 91.4% of held-out wells, though typical intervals spanned two orders of magnitude. An exploratory combined arsenic-iron-manganese risk score was dominated by manganese and did not usefully separate wells. Applying the WHO guideline ($10 \mu\text{g/L}$) instead of the national standard raised the exceeding share from 24.9% to 42.1%. The results characterize model behavior on 1998–1999 data and cannot certify the safety of any well today.

Keywords: arsenic, Bangladesh, groundwater, gradient boosting, spatial cross-validation, explainable AI, conformal prediction

Introduction: The Bangladesh Arsenic Problem

The World Health Organization has characterized chronic arsenic exposure from Bangladeshi well water as the largest mass poisoning of a population in history (Smith et al., 2000). Beginning in the early 1970s, national and international programs promoted shallow tube wells as a safe alternative to bacterially contaminated surface water. The wells succeeded at reducing diarrheal disease, but by the mid-1990s it became clear that many of them drew water from Holocene alluvial and deltaic sediments naturally rich in arsenic, released into groundwater through the microbially mediated reduction of iron-oxyhydroxides under anoxic conditions (Smedley & Kinniburgh, 2002). Estimates of the population exposed above the national standard of $50 \mu\text{g/L}$ have ranged from about 21 million shortly after the problem was first documented (Smith et al., 2000) to 35–77 million across surveys conducted between 2000 and 2010 (Flanagan et al., 2012). Chronic exposure has been prospectively linked to increased all-cause, cancer, and cardiovascular mortality in Bangladeshi cohorts (Argos et al., 2010). The crisis is compounded by scale: nationwide field-kit screening between 2000 and 2003 covered only 4.94 million tube wells, of which 29% exceeded the national standard, and testing of the many millions of wells drilled since has been far less complete (Flanagan et al., 2012).

This creates a specific, practical problem. Because arsenic is colorless, odorless, and tasteless, a well cannot be judged safe by inspection, and testing every well with laboratory methods (or even field kits) at national scale is slow and expensive to repeat as aquifer conditions and mitigation change. This motivates sustained interest in whether machine-learning models trained on existing survey data can extract dependable, generalizable, explainable risk patterns – for example to help prioritize where new testing or mitigation (well switching, deep wells, filters) should happen first (Flanagan et al., 2012). Doing so responsibly requires answering questions often skipped in exploratory work: whether accuracy survives testing on geographically new areas; what a model actually bases predictions on; how confident individual predictions are; and whether combining contaminants into one score is trustworthy.

This study analyzes the 1998–1999 National Hydrochemical Survey of Bangladesh (British Geological Survey [BGS] & Department of Public Health Engineering [DPHE], 2001), treating it strictly as a historical dataset for methodological analysis rather than a present-day safety assessment. It asks: (a) which measured variables are associated with arsenic exceedance; (b) how much reported accuracy depends on random versus spatially aware validation, and how much that estimate itself varies with which areas are withheld; (c) what the best model’s predictions depend on, and how uncertain individual predictions are; (d) whether a combined multi-contaminant score is useful; and (e) how conclusions change

under a stricter guideline. Since 1999, large-scale mitigation means present conditions at the same coordinates may differ substantially from the 1998–1999 readings (Flanagan et al., 2012); this distinction is maintained throughout.

Method

Data and Cleaning

The data are `NationalSurveyData.csv`, a working copy of the original National Hydrochemical Survey (BGS & DPHE, 2001) obtained via a Kaggle mirror of the published survey: 3,534 wells with 14 descriptive columns (location, depth, year built, well type) and 20 measured elements. Figure 1 shows the geographic spread of sampled wells and their arsenic readings: coverage is dense across the center and south but thinner in parts of the north-east, and the highest readings (yellow-green) cluster across a broad belt of the south-center and south-west, while much of the north and north-west reads low (dark purple).



Figure 1: Sampled wells (left) and the same wells shaded by arsenic reading on a log scale (right), 1998–99 survey.

No duplicate wells or out-of-country coordinates were found; four implausible depths were set to missing. Readings below a laboratory detection limit (e.g., “<0.5”) were replaced with half the reporting limit, with a flag retained for substituted values; this affected 31.3% of arsenic, 44.1% of aluminum, 28.2% of phosphorus, and 23.9% of sulfate readings, but under 1% of iron and manganese. Figure 2 shows the resulting distributions: the tall bar at the left of the arsenic panel is exactly this substituted mass, iron is bimodal (two loose well populations), and manganese is unimodal. Figure 3 plots the same three elements against well depth: the highest readings concentrate in shallow wells, and the small cluster deeper than ~250 m trends lower, though scatter at every depth is wide.

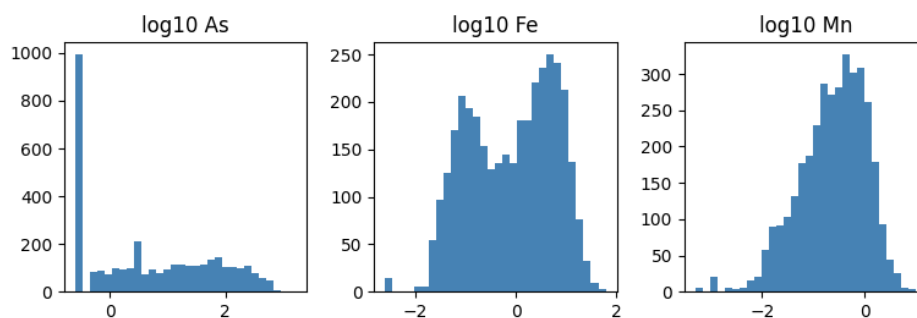


Figure 2: Log₁₀ distributions of arsenic, iron, and manganese readings.

Exceedance targets used Bangladesh’s Schedule 3B standard (Government of Bangladesh, 1997) – 50 $\mu\text{g/L}$ arsenic, 1.0 mg/L iron, 0.1 mg/L manganese; the WHO guideline of 10 $\mu\text{g/L}$ (WHO, 2011) was kept for sensitivity comparison. Table 1 gives exceedance rates, consistent with the range of contamination previously reported nationally (Smedley & Kinniburgh, 2002).

Prediction Scenarios

Scenario A (explanatory) combines a sample’s full measured chemistry with well-construction details, asking what accounts for arsenic already measured; it cannot screen an untested well. **Scenario B** (screening) uses only what is knowable before a visit – coordinates, division, district, depth, year, type – the only setup that could characterize an unsampled well.

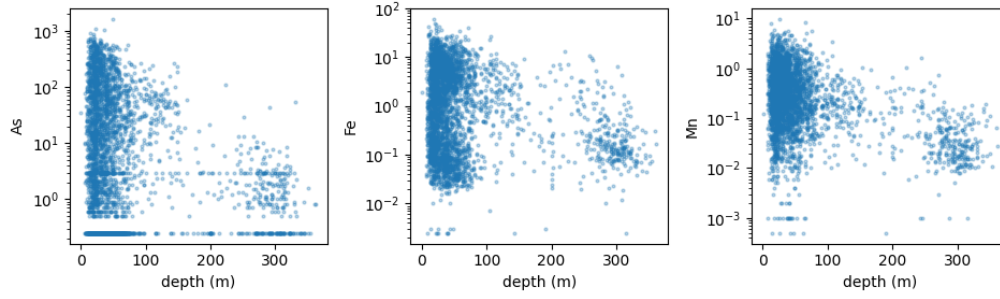


Figure 3: Arsenic, iron, and manganese vs. well depth (log scale). Descriptive only.

Table 1: Guideline Values and Observed Exceedance Rates

Element	Bangladesh guideline	WHO comparison	Wells above (share)
Arsenic	50 $\mu\text{g/L}$	10 $\mu\text{g/L}$	881 / 3,534 (24.9%)
Iron	1.0 mg/L	–	1,806 / 3,530 (51.2%)
Manganese	0.1 mg/L	–	2,596 / 3,530 (73.5%)

Classification, regression, SHAP, and intervals use Scenario A; the multi-contaminant score uses Scenario B; the two are never combined.

Models

Two forms of the arsenic question were modeled: classification (exceeds 50 $\mu\text{g/L}$ or not) and regression ($\log_{10}$ concentration, which keeps the 49-vs-51 $\mu\text{g/L}$ distinction a binary label discards). Baselines: a dummy classifier at the base rate and logistic regression for classification; a median predictor and ridge regression for regression. Five learning algorithms were then compared:

- **Random forest** – an ensemble of trees, each trained on a bootstrap sample with a random subset of variables at each split, averaged across trees. Robust to irrelevant variables; a natural tree-ensemble baseline.
- **XGBoost, LightGBM, CatBoost** – three gradient-boosting implementations. Each builds trees sequentially, with every new tree fit to the residual errors of the ensemble so far, unlike random forest’s independent trees. They differ in tree-growth strategy and categorical handling (CatBoost encodes categories natively; XGBoost/LightGBM used one-hot here) and in speed/regularization engineering. All three typically outperform linear models and single trees on tabular data.
- **Multilayer perceptron (MLP)** – a small feed-forward network, one hidden layer of 32 units, testing whether a neural approach adds anything over tree ensembles at this dataset’s size.

Preprocessing (imputation, scaling, one-hot encoding) was fit only on training folds inside a pipeline to prevent leakage. The four tree-based models were tuned via randomized search over 12 hyperparameter settings under cross-validation; the MLP used fixed defaults and was not tuned to the same extent, likely explaining part of its lower, more variable scores.

Validation Design

Groundwater chemistry is spatially autocorrelated, so a random split can let a model see a near-duplicate of a test well during training (geographic leakage), inflating accuracy. Three schemes were compared: (1) a random, stratified split; (2) a single spatial-block holdout – 209 grid blocks of 0.25° (≈ 25 km), withholding whole blocks until $\sim 20\%$ of wells (717 of 3,534) were held out; and (3) the same holdout repeated eight times with different withheld blocks, to quantify how much the estimate itself depends on which areas are excluded. Hyperparameters were tuned once on the reference spatial split and reused across repeats. A block-size check re-ran the repeated holdout at half and double block width for the best model.

Interpretation and Uncertainty

Model behavior was examined with SHAP (Lundberg & Lee, 2017), attributing each prediction to the variables that produced it, applied to the CatBoost classifier on the 717 spatial holdout wells. For regression, conformalized quantile regression (Romano et al., 2019) produced 90% prediction intervals from two gradient-boosted quantile models (5th/95th percentile), corrected on a separate spatially held-out calibration set (567 wells) so coverage was measured, not assumed; a

bootstrap interval was computed for comparison. For the multi-contaminant score, out-of-fold Scenario B logistic regressions (spatially blocked cross-validation) estimated exceedance probability for arsenic, iron, and manganese, combined as $1 - \prod_i (1 - p_i)$ – an exploratory construct assuming independence between contaminants, which is unlikely to hold exactly.

Results

Classification of Arsenic Exceedance

Table 2 shows the four tree-based models performing similarly well, with repeated-spatial mean ROC-AUC of 0.950–0.953. The MLP scored lower and more variably ($M = 0.893$, $SD = 0.039$); logistic regression reached 0.874; the dummy baseline scored 0.500. Averaged across the five main models, random-split ROC-AUC exceeded the repeated-spatial mean by only 0.001 – the leakage penalty spatial validation is meant to reveal essentially did not appear, possibly because Scenario A models never see coordinates directly. Figure 4 shows why a mean-with-spread is the right way to report this: across the eight spatial repeats, CatBoost ranged from 0.935 to 0.962 and the MLP from 0.802 to 0.923 (visible as its low outlier), a spread larger than the gap between the top four models. The block-size check gave mean ROC-AUC of 0.950 (700 fine blocks), 0.950 (209 main blocks), and 0.945 (68 coarse blocks) for the best model, so the headline result is not an artifact of grid choice. At a fixed flagging rate, CatBoost reached precision = 0.814 and recall = 0.774 on the single spatial split.

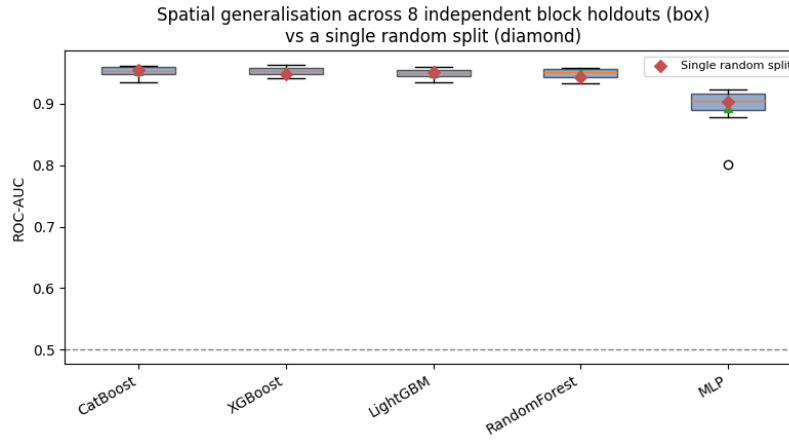


Figure 4: ROC-AUC across eight spatial block holdouts (box) vs. a single random split (diamond), by model.

Table 2: Classification and Regression Performance (Scenario A)

Model	Classification ROC-AUC		Regression RMSE ($\log_{10}$)		PR-AUC
	Single spatial	Repeated spatial M (SD)	Single spatial	Repeated spatial M (SD)	
Dummy / median	0.500	–	1.121	–	0.259
Logistic / ridge	0.874	–	0.827	–	0.719
Random forest	0.947	0.950 (0.009)	0.566	0.549 (0.022)	0.868
XGBoost	0.950	0.953 (0.008)	0.542	0.534 (0.023)	0.872
LightGBM	0.952	0.950 (0.009)	0.551	0.537 (0.022)	0.877
CatBoost	0.953	0.953 (0.009)	0.537	0.523 (0.020)	0.877
MLP	0.906	0.893 (0.039)	0.573	0.591 (0.026)	0.786

Note. Single-spatial split has 717 test wells; RMSE on base-10 log-arsenic; PR-AUC on single spatial split.

Regression on Log-Transformed Arsenic

CatBoost was the best regressor (Table 2), with single-split $R^2 = 0.768$ and Spearman correlation = 0.859, against $R^2 = 0.450$ for ridge regression. An RMSE near 0.5 on the log scale corresponds to roughly a three-fold typical error – useful for ranking wells, not a substitute for measurement.

What the Model Relies On

Figure 5 shows the SHAP summary for the CatBoost classifier: each row is a variable ranked by influence, each dot a well, colored by that well's value of the variable (blue = low, red = high), positioned by its push toward (right) or away from (left) predicted exceedance. Phosphorus (mean $|\text{SHAP}| = 1.17$), iron (0.99), and strontium (0.76) were most influential, followed by silicon (0.49), potassium (0.47), and sulfate (0.42); a second tier – sodium (0.22), well depth (0.22), well type STW (0.20), and smaller contributions from magnesium, calcium, manganese, barium, and aluminum – mattered less, and year built and the remaining well-type indicators barely registered. High phosphorus, iron, strontium, or potassium pushed toward exceedance; high silicon, sulfate, sodium, and greater depth pushed away, consistent with Figure 3. These are associations the model uses, not causes: iron may simply track the same reducing conditions that release arsenic rather than driving it. SHAP was computed for one model on one split.

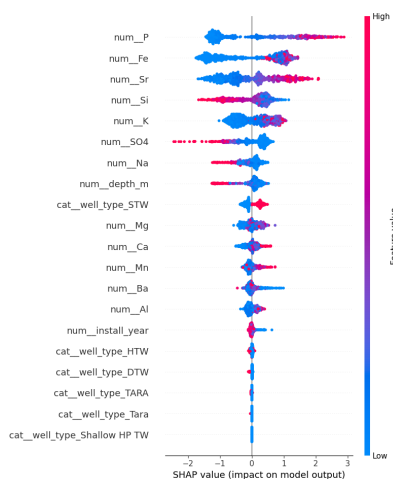


Figure 5: SHAP summary, CatBoost arsenic classifier, held-out spatial test wells. “num__” = numeric variable; “cat__well_type_” = well type.

Prediction Intervals

On the 717 held-out wells, the conformal 90% interval covered 91.4% of observed values (bootstrap comparison: 90.0%), but typical intervals were wide: median width 2.00 log units, i.e., a typical interval's upper bound is ~ 100 times its lower bound (range 0.66–3.48). Figure 6 shows this is not uniform: intervals were narrower for wells predicted low-risk and wider for wells predicted high-risk, so the model is least certain exactly where certainty matters most, and for roughly half the wells the lower bound sits at the below-detection floor visible as the flat line of points at the bottom.

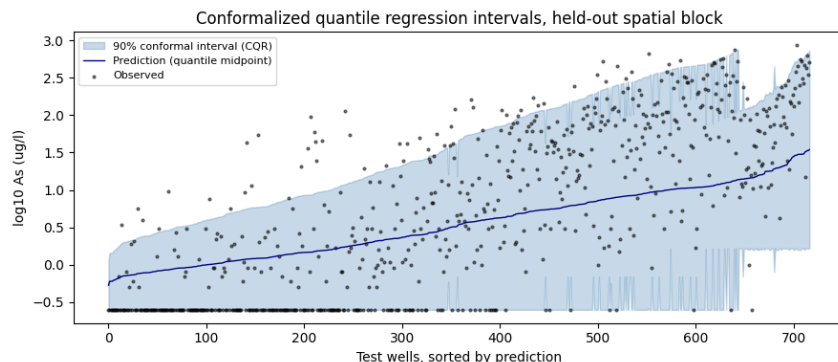


Figure 6: Predicted arsenic and 90% conformal intervals, 717 held-out wells sorted by prediction.

Multi-Contaminant Score and Guideline Sensitivity

Scenario B screening models (location and well details only) reached ROC-AUC of 0.846 (arsenic), 0.755 (manganese), and 0.719 (iron) – better than chance but well below the ~ 0.95 of chemistry-based Scenario A models. The combined score was not useful: mean 0.87, median 0.95, so under 0.2/0.5 cut-points, 3,310 of 3,534 wells fell into a single “high”

bucket. Manganese (exceeding in 73.5% of wells) drove this: removing it dropped the mean to 0.59, while removing iron or arsenic changed it only to 0.79 or 0.85. Substituting the WHO guideline for the national standard raised the exceeding share from 24.9% to 42.1% nationally; 16.7% of held-out wells fell between the two thresholds. Ranking quality was similar under both (ROC-AUC 0.885 national vs. 0.913 WHO).

Spatial Patterns

Both maps below describe 1998–99 only, not present-day conditions. Figure 7 compares sampled wells, observed exceedance, and the CatBoost model’s out-of-fold predicted probability: the model reproduces the observed south-center/south-west belt of contamination well in areas it was not trained on, with intermediate probabilities correctly appearing in the transition zones. Figure 8 (left) shows the exploratory composite score is high almost everywhere – as the manganese analysis above predicts – with a visible low-scoring cluster in the south; (right) shows conformal interval width for held-out wells only, with narrower (darker) intervals concentrated in the north-west and wider (lighter) ones in the center and south, matching Figure 6.

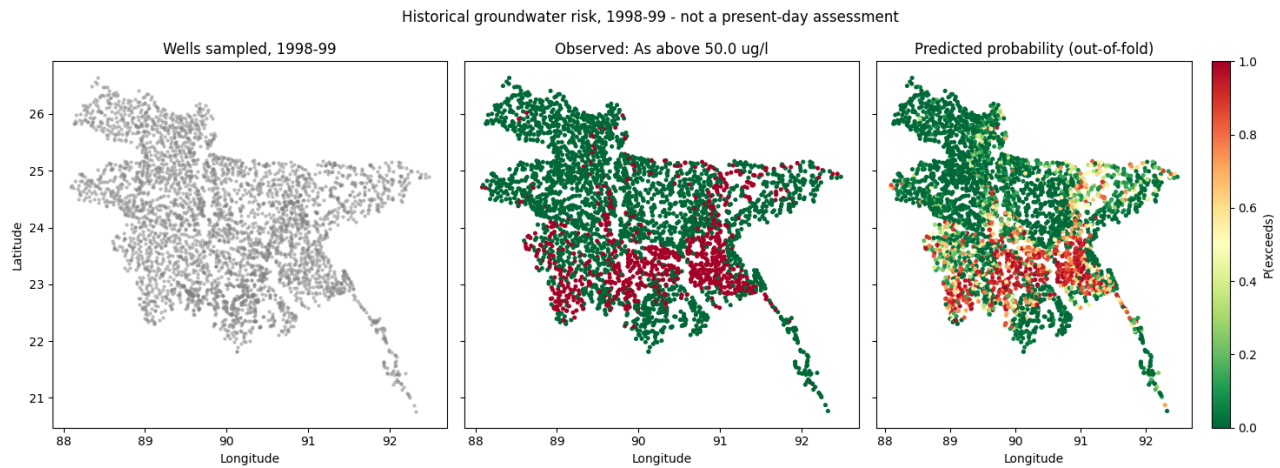


Figure 7: Sampled wells, observed arsenic exceedance, and out-of-fold predicted probability of exceedance, 1998–99.

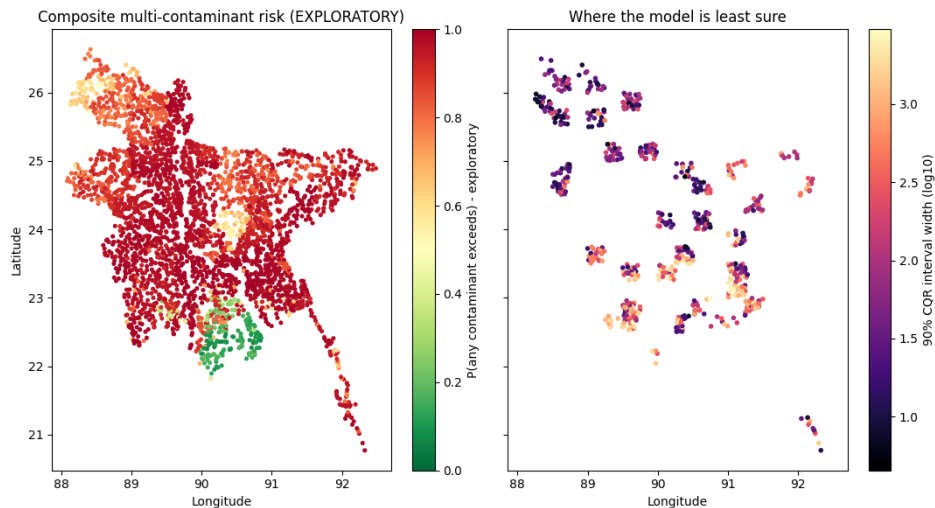


Figure 8: Exploratory composite multi-contaminant risk (left) and 90% interval width for held-out wells only (right).

Discussion

Given the scale of Bangladesh’s arsenic problem – tens of millions of people drinking above-guideline water from millions of largely untested wells (Flanagan et al., 2012; Smith et al., 2000) – a validated, explainable, honestly uncertain screening tool has obvious appeal. This analysis suggests such tools are promising but not sufficient alone. Tree-based models (especially CatBoost and XGBoost) ranked historically surveyed wells with high, spatially robust accuracy: the

anticipated drop from random to spatial validation did not materialize, and the four leading algorithms were statistically indistinguishable given the variation caused by which areas were withheld. SHAP made the classifier’s reasoning inspectable (phosphorus, iron, strontium most influential) without claiming causal mechanism. The conformal intervals met their coverage target but were wide enough to temper, not replace, laboratory confirmation. The combined multi-contaminant score failed operationally because manganese exceedance is nearly ubiquitous, illustrating how a well-intentioned composite index can mislead. The choice between the national standard and the WHO guideline roughly determines whether 25% or 42% of wells count as unsafe – not a minor detail for mitigation planning.

Limitations. The survey is 27 years old; large mitigation since 1999 means present conditions may differ substantially, so nothing here certifies any well safe today. Roughly 31% of arsenic readings were below-detection substitutions; units were assumed; hyperparameters were tuned once and reused across repeats; SHAP and the interval method were each evaluated on one split; the multi-contaminant score assumes contaminant independence unlikely to hold; and the data lack geology, sediment age, and land-use variables that likely explain some unmodeled spatial structure.

Future directions. Applying this pipeline to newer groundwater measurements, adding geological/land-use/hydrological covariates, fully nested spatial tuning, and field validation (testing wells where the model is confident and checking the result) would be natural next steps before any operational use.

Conclusion

Applied carefully, machine learning can extract dependable, explainable, honestly uncertain patterns from historical groundwater surveys of the kind that document Bangladesh’s arsenic crisis, and spatial validation shows at least some of this accuracy is not an artifact of geographic leakage. But the exercise also shows how quickly conclusions shift with analytic choices – which guideline is applied, how contaminants are combined, which areas are withheld – and how far a model’s ranking ability is from a guarantee about any single well. In a country where millions of wells remain untested, such tools may help prioritize where to look next, but the decision that a specific well is safe to drink from remains a matter for direct measurement.

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